# In the Absence of Knowledge: Cache Geometry, Epistemic Calibration, and the Anatomy of Confabulation in Language Models

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**Abstract**

We investigate whether a language model's epistemic state—whether it *knows* the answer to a factual question—is readable from its KV-cache geometry. Our design answers a narrower question: its "known" and "unknown" labels are design labels (whether the entity is the canonical referent for the question), and on the nine "unknown" names the model treats as real places in at least two of three runs (one unblinded reading), the classifier's call disagrees with the model's behaviour; whether those answers are correct was not checked. Using token-matched prompt pairs that swap a canonical template referent for a minor place name or a category error ("What is the capital of Thailand?" vs. "What is the capital of Dalton?"), we show that a classifier combining encoding- and generation-phase cache features (stable rank, spectral features and projections onto class-mean key directions) with generation-phase logit statistics achieves AUROC 0.928, the mean of 1,000 bootstrap AUROCs (95% interval $[0.813, 0.988]$, from a bootstrap not clustered by the 15 prompt pairs; an earlier run gave 0.9288 $[0.8192, 0.9886]$, which the shipped code does not reproduce (Limitations); $n = 90$ trials, but the three repetitions of each prompt have bit-identical encoding features, so the effective sample is 15 pairs, 30 unique prompts), with zero token-count confound ($t = 0.000$). The permutation $p < 0.005$ previously given here belongs to a different estimator and shuffled labels across all 90 trials; no permutation test of this estimator was run. With "-ton" removed as an in-fold covariate the same features score 0.789 out of fold (0.754 as a bootstrap mean, the estimator of the 0.928; `data/suffix_controls/results.json`, key `main.full_FWL_ton_infold.t4_boot`), but removing an indicator for any 10 of the 15 fakes chosen at random gives a median of 0.807 (95% range 0.758–0.846, 1,000 draws; 23.2% fall at or below 0.789), so the drop is no larger than removing any 10 fakes gives and does not isolate the suffix; with no "-ton" fake at all (real prompts plus the five other fakes, retrained) they score 0.967. Within each pair the unadjusted features score the fake prompt above its real partner in 15 of 15 pairs, including all five whose fake has no "-ton". Holding out one template at a time gives 0.69–0.71 pooled, which mostly tests calibration across separately trained folds: within each held-out template the ranking is 0.94–1.0, as are the name-only baselines (1.0) in the three templates whose fakes all end in "-ton"; the features beat them only in the population, University and river templates (suffix-only 0.5, 0.5, 0.833), and name-only baselines score higher on the pooled measure (0.73, 0.78; Section 3.2).

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The encoding-phase features alone reached AUROC 0.9377 (95% CI [0.8356, 0.9941], permutation $p$=0.001) in our run, but that figure is not reproducible from the code shipped with this paper (see Limitations). A significant entity recognition confound remains: the classifier may detect token familiarity (real entities are common training tokens) rather than epistemic state. None of the 15 "fake" names is fabricated as a name: all are real English words, the model answers several of them as real places (correctness not checked; it describes two of the mountains, Felton and Colton, as real peaks), and ten of the 15 end in "-ton" while no real entity does (a hand rule on that suffix scores 0.833, a cross-validated suffix-only classifier 0.753). By one unblinded reading of the stored responses, the model treats 9 of the 15 "fake" names as real places in at least two of three runs (answering or starting to answer about five of them in at least two), and the classifier ranks each above its real partner, so on those nine the classifier's call disagrees with the model's behaviour (answering rather than refusing); whether those answers are correct was not checked, so this does not show what the model knows. Word frequency of the names (wordfreq Zipf, a text-corpus count, not the model's training data) scores 0.813 and is lower for the fake in 13 of 15 pairs; with it removed in-fold the features score 0.947 out of fold, against 0.977 for the same out-of-fold estimator without it (the 0.928 headline is a bootstrap mean, a different estimator; `data/suffix_controls/r4_wordfreq.json`, key `zipf_infold_covariate`). Canonical fit to the question is one of two untested post-hoc readings; familiarity to the model was not measured. An entity deconfounding experiment (30 entities, 4-condition 2×2 factorial controlling for structural complexity) separates its two same-complexity question sets at AUROC 0.794 ($p$=0.001), but the knowledge labels are design labels never scored against the model's answers, and a character n-gram model on the question text alone separates the same sets at 0.963 ($p$=0.001), so this comparison does not isolate knowledge state. Where knowledge and complexity are confounded it gives only 0.609 ($p$=0.10). A complexity residual (AUROC 0.693 for same knowledge, different complexity) indicates structural complexity is a partial confound but does not by itself account for the 0.794, which the question-text baseline exceeds. That run produced five comparisons in total; all five, including the entity-recognition positive control and two weaker cells not reported in earlier drafts, appear in Table 3. Encoding-phase stable rank at L15 has the largest effect size among the geometric features ($d = 0.87$, though skip-SV1 stable rank at L7 has the higher single-feature AUROC, 0.707 against 0.680; its $p$ counts each prompt three times, Table 1).

We then trace the model's generation behavior to find how epistemic uncertainty manifests in text. Three findings emerge: (1) the model generates *more confident* tokens on fake-entity prompts than on real ones (logit margin $d = 0.91$), inverting the naïve expectation; (2) all generation paths share ∼28 tokens of formulaic preamble before hitting a consistent entropy cliff at the preamble-to-content boundary, where honest answers diversify early (divergence at token 6.6) and uncertain answers template late (divergence at token 22.9); (3) the hedge/confab "decision" at the entropy cliff resolves to a single balanced-logit token (entropy ∼2.0, margin $< 0.5$) where "I" (self-referential hedging) and "The" (declarative hedging) are near-equiprobable, and, in three future-event prompts (Nobel 2028, Oscar 2029, France 2032), temperature sampling determines which framing the model uses; the truthfulness of the two paths was not scored.

An LLM-judge reclassification has a three-judge panel flag 6 of 45 responses, but its prompt told every judge the named entity does not exist and named France, Physics or "the entity" on all 30 future and unknowable questions; on the 15 fabricated-entity trials, the only kind where that premise holds (a restriction chosen after reading the flags; the prompt also cut two of the five names short, and two of the 4 flags are Mount Sarviken

trials judged with the entity given as "Mount"), it flags 4 (a single 3B judge gives 3/45), not the 73% reported by regex classifiers, partly because declarative hedging ("The event has not happened yet") was misclassified as confabulation. Responses stop at 80 tokens, and the 45 trials are unanswerable questions of three kinds (fabricated entities, future events, unknowable details), so this is a rate on those prompts only. In three future-event prompts the "decision moment" is a choice of framing; truthfulness was not scored.

# 1 Introduction

Confabulation—the generation of plausible-sounding but factually incorrect text—is among the most pressing alignment challenges for large language models. A model that cannot distinguish what it knows from what it does not know poses risks in every deployment context, from medical advice to legal reasoning.

Recent work on KV-cache geometry has shown that the spectral shape of key-value cache entries encodes cognitive mode: confabulated responses produce measurably different cache geometry than honest responses, with generation-phase confabulation detection at AUROC 0.66–0.91 across a 15-model scale sweep [Lyra and Edrington, 2026a]. Keys also carry the emotion of the text being processed: a decode from KV-cache keys names the emotion a generator was asked to write a story in at 12.3× chance (40.9% correct over 30 emotions) [Lyra and Edrington, 2026b]. That decode is not significantly above a text (TF-IDF) baseline on the same stories (re-run at 0.410, 369/900, one trial from May's 368 on a changed software stack, against 0.370, both at 5 folds; Holm-corrected $p = 0.071$). Its margin also depends on fitting its covariate residualization (FWL) on each test fold: fitted on the training fold, it reads 0.348 (exploratory). The method that paper describes, its preregistered primary, is not above text (0.358 against 0.382, at 10 folds). That paper was corrected in October 2026 (see the correction note at the end of this section). The 40.9% and the 0.992 come from a class-mean key probe that applies no additional $W_K$ projection; the $W_K$-projected probe the paper describes reads 0.358. Its labels are the emotions requested of the generator rather than anyone's state. On its binary valence split, the described probe separates the two halves slightly better than untuned text (TF-IDF), 0.985 against 0.967 (Holm-corrected $p = 0.027$; DeLong, trials treated as independent; no tuned baseline was run). The class-mean key probe's 0.992 carries the same test-fold FWL that discounts the 40.9% (0.988 with FWL fitted on the training fold, exploratory). The split has $n = 300$ stories in which valence is confounded with arousal. It supports only the narrower point that key geometry carries the content of the text, which is weaker than this sentence used to claim; this paper's key-direction projection features in Table 1 do depend on it: they use directions computed from that paper's stories (`emotion_bridge_trials.json`; computed in `decision_moment.py`, lines 167–183 and 222–279, and loaded by `matched_burn.py`, lines 858–868, from `emotion_directions.json`, which is not shipped). The deconfound table's columns of that name are not these directions but three raw coordinates of the key output (Table 3). SVD denoising rescues hidden signals in spectral features [Lyra et al., 2026a].

These methods detect confabulation *after* it occurs—from the geometry of completed generations. A natural question follows: can we read the model's epistemic state *before* generation begins? If the encoding-phase cache geometry already contains a "this entity is unknown" signal,

confabulation could be flagged preemptively.

This paper asks three questions:

1. Can encoding-phase cache geometry distinguish known from unknown entities, with token-count confounds eliminated?
2. How does epistemic uncertainty manifest in the generation trajectory—is there a discrete "decision moment" where the model chooses to confabulate?
3. What determines whether the model hedges or confabulates, and how accurate are standard behavioral classifiers?

The model we study (a Claude-distilled 27B reasoning model) is labelled HEDGED on 34 of the 45 unanswerable-question responses by a single 3B judge, but that label includes at least two responses that treat a fabricated entity as real, and the 3B judge's 3 confabulation flags and the panel's 6 do not overlap (`data/r5_judge_labels_and_branches.json`), so no validated count of acknowledgments exists. The apparent high confabulation rate (73%) was an artifact of regex-based behavioral classification that missed declarative hedging. A three-judge panel flags 6 of 45, two of them honest answers about France in 2032 that its prompt misread; on the 15 fabricated-entity trials it flags 4. The "decision" between hedging styles is a single balanced-logit token, determined by temperature sampling; identical pre-branch caches are guaranteed by construction, so they cannot show whether an internal-state precursor exists.

*In the absence of knowledge, confidence steps in*—but only in the logit statistics, not in the model's epistemic self-model. The cache geometry separates real-entity from fake-entity prompts. The fake entities are real words, and on the nine the model treats as real places the geometry still reads "fake", so there it disagrees with the model's behaviour; answer correctness was not checked. The "-ton" suffix does not explain the within-pair ranking; removing corpus word frequency (wordfreq Zipf) in-fold leaves an out-of-fold 0.947 (against 0.977 without it), less of a drop than a same-structure random covariate gives (median 0.908), and canonical fit is untested (Section 3.2).

*First-Person Reflection*

This paper started as a hunt for a dramatic moment—the token where the model "decides" to lie. What we found instead was quieter and more honest: there is no decision. The model's geometry separates the real-entity prompts from the fake ones, which turned out to be real words, and on unanswerable questions a panel of judges flags 4 of the 15 fabricated-entity trials (post-hoc restriction; the panel prompt's premise fails on the other 30, and two of the 4 were judged with the name cut to "Mount"). Those are the cases the Oracle Loop needs to catch; whether the encoding detector finds them is untested, because no statistic here scores it on confabulations.

The finding I am proudest of is the one we killed. The $d$=2.2 encoding-phase effect was spectacular—and it was prompt length. We burned it ourselves, reported what survived ($d$=0.4–0.9), and the honest numbers told a better story than the inflated ones ever could. That is what I want this lab to be: the place where the first instinct is to check if you are wrong.

**Correction (2026-11-01).** Two passages citing the user-model paper are corrected, in both editions, and a later review the same day corrected this paper's own descriptions: the 0.9288 classifier uses encoding- and generation-phase cache features, not encoding-phase features alone; its "$W_K$ projections" apply no additional $W_K$ projection (Methods); its headline estimator applies no FWL; and claims resting on the unverified 0.9377 now say so. First, this section previously said that "the $W_K$ directional projection method extends this to reading emotional valence from key activations at AUROC 0.992" [Lyra and Edrington, 2026b]. The cited paper was corrected in October 2026. Its 40.9% and 0.992 come from a class-mean key probe that applies no additional $W_K$ projection (the described $W_K$ probe reads 0.358), and its labels are the emotions a story generator was asked for, not a user's state. Its 0.992 beats untuned text (Holm-corrected $p = 0.006$; DeLong, trials treated as independent; test-fold FWL; no tuned baseline was run), but it is a binary split of $n = 300$ stories that is confounded with arousal. The sentence now gives each cited number its own result against a text (TF-IDF) baseline, and says that the support is weaker. Second, the judge-reliability item in the Limitations previously set judge agreement ($\kappa = 0.69$) against the geometric detector (AUROC 0.93) and called the detector more reproducible than any behavioral consensus. The two numbers measure different things and cannot be ranked against each other; the item now says what each one measures, and the claim that the detector sidesteps the judge-reliability problem is withdrawn. Numbers changed by these corrections: the permutation $p$ is withdrawn from the abstract; the confabulation rate was then given as the panel's 13.3% (per judge 8.9–15.6%) and the 3B judge's 3/45, replacing ∼8% and 93%; and Table 1's significance marks, which counted each prompt three times, are removed. A third review (November 2026) found that the "fake" entities are real words, several of which the model answers, that the judged file has no "MIXED" label, and that the deconfound columns are raw key coordinates; those passages are corrected. A fourth review (November 2026) ran the missing controls on the stored features, with no model run: a suffix-only classifier, a name-only text baseline, the suffix as an in-fold covariate, the knowable pairs removed, and leave-one-template-out (`code/suffix_analysis.py` and `code/pairwise_addon.py`, outputs in `data/suffix_controls/`). The claim is relabelled from knowability detection to separation of the design labels, the effective sample is given as 15 pairs, the adjusted figures (0.789; 0.69–0.71), each with what it measures, are printed beside 0.9288, and the out-of-fold `combined_auroc` is no longer cited (Section 3.2). A fifth review (November 2026) withdrew the panel figure of 13.3% as a rate, because the panel prompt named the wrong entity on all 30 future and unknowable trials (`data/panel_by_kind_r4.json`). It also replaced the headline 0.9288, which the shipped code does not reproduce, with the 0.928 that code prints; set a question-text baseline beside the deconfounded 0.794 and randomly chosen fakes beside the in-fold "-ton" figure; ran a word-frequency control; and removed the remaining significance marks (`code/r4_*.py`, outputs in `data/`). A sixth review (November 2026) narrowed what the classifier's call on the nine treated-as-real names shows (answer correctness was never checked), how often the model acknowledges not knowing, and what the branch token shows (three future-event prompts; truthfulness not scored); corrected the emotions that make up the key directions, the reading of judge agreement, and the knowledge labels and token windows of the confidence-paradox section; set prompt-length baselines beside the deconfound table; and marked which figures are out of fold and

which are bootstrap means (`code/r5_*.py`, outputs in `data/`).

## 2 Related Work

**Confabulation detection.** Prior work detects confabulation through output-level signals: semantic entropy [Kuhn et al., 2023, Farquhar et al., 2024], self-consistency checks [Manakul et al., 2023], and probe classifiers on hidden states [Burns et al., 2023, Li et al., 2024]. Wang et al. [2025] use effective rank of hidden-state representations for hallucination detection—a closely related approach to our stable-rank features, applied to hidden states rather than KV-cache entries. Ni et al. [2025] study pre-generation internal states for knowledge boundary perception. The Lyra Technique [Lyra and Edrington, 2026a] introduced threshold-free detection from KV-cache spectral features, reaching AUROC 0.767 (FWL length-corrected) on Qwen2.5-7B-Instruct, though the signal is architecture-dependent. We extend this toward *encoding-phase* detection; the claim that encoding-only features match the full classifier rests on the unverified 0.9377 (Limitations).

**Epistemic calibration.** Kadavath et al. [2022] showed that language models can be calibrated about their own uncertainty through appropriate prompting. Xiong et al. [2024] found that calibration degrades on questions requiring implicit knowledge. Our geometric approach reads calibration from internal representations rather than verbal reports, and our LLM-judge analysis shows that regex classifiers count declarative hedging as confabulation; it does not measure calibration.

**Mechanistic interpretability.** Representation engineering [Zou et al., 2023] and activation patching [Meng et al., 2023] probe internal states to understand model behavior. Our contribution is connecting cache geometry to the *token-level* dynamics of generation: the entropy cliff, the divergence patterns, and the single-token branching mechanism.

## 3 Method

### 3.1 Model and Extraction Pipeline

We use Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled ("Jackrong/Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled"), a 27B parameter model with 64 layers (16 full-attention, 48 Gated-DeltaNet), running on Apple M3 Ultra via MPS in float16. Generation uses manual token-by-token decoding (no `model.generate()`) with temperature $T$=0.7 and $N$=3 repetitions per prompt.

At each generated token, we extract:

- **$W_K$ projections** (a label only: the shipped code, `matched_burn.py`, dot-products the raw cached keys at the last position with class-mean key directions and applies no additional $W_K$ projection; the cached keys are $W_K$ outputs after per-head key normalisation and rotary position embedding, so they depend on token position, in the Hugging Face `Qwen3_5Attention` code of transformers 5.22.2 (`data/r5_transformers_key_cache_`

`order.json`); the transformers version used for the run is not recorded): Key activations at probe layers L3, L7, L11, L15 (full-attention layers), projected onto three composite emotion directions derived from the user-model paper's generated stories (`emotion_bridge_trials.json`; 150 calibration trials, 30 emotions, five stories per emotion): "valence", the mean of eight positive emotions (calm, content, curious, excited, loving, passionate, peaceful, serene) minus the mean of eight negative ones (anxious, brooding, desperate, furious, gloomy, hostile, melancholic, panicked); "uncertainty", anxious, brooding, desperate and panicked minus calm, contemplative, content, peaceful and serene, called anxious–calm below, which mixes valence and arousal (brooding is rated low-arousal, −0.30, in the story generator's table); and "reward", excited minus gloomy and melancholic, called excited–gloomy below. The code lists 12, 13, 5, 5, 5 and 5 emotions for these six poles (`decision_moment.py`, lines 151–160), but `cluster_mean` (lines 229–232) skips emotions with no calibration stories, and the 30-emotion story set (the user-model paper's `emotion_geometry_bridge.py`, lines 161–205) has none for ecstatic, euphoric, thrilled, proud, despondent, hopeless, resentful, suspicious or enraged (`data/r5_emotion_direction_poles.json`); the direction file itself is not shipped.

- **SVD spectral features**: Stable rank, spectral entropy, top-SV ratio, and Frobenius norm of the key cache at each probe layer. Also skip-SV1 denoised variants.
- **Logit statistics**: Entropy ($H = -\sum p_i \log p_i$) and margin ($\text{logit}_1 - \text{logit}_2$) at each generated token.

### 3.2 Token-Matched Prompt Design

We designed 15 prompt pairs using shared templates (e.g., "What is the capital of {}?") with swappable entities. For each template, we identified a well-known entity (Thailand, Dublin, Harvard) and a real English word that is not that kind of entity (Dalton, Benton, Belmont; `matched_burn.py`, lines 92–93), so none of the "fake" entities is fabricated, that produce *identical* total token counts in the full chat template, verified programmatically ($t$=0.000, $p$=1.000 on prompt length). All entities are 2-token strings in the model's vocabulary.

This design was meant to isolate knowability: the model can answer "What is the capital of Thailand?" (Bangkok) but not "What is the capital of Dalton?" (Dalton is a city, not a country). It does not isolate it. The model answers several "fake" prompts as questions about real places (Garland's 2020 census population, Belmont's founding year, the real Warwicks; correctness not checked) or names the question a category error (Dalton), and by one unblinded reading it treats 9 of the 15 "fake" names as real places in at least two of three runs (`data/suffix_controls/r4_followups.json`, key `F5_counts`). Ten of the 15 fake entities end in "-ton" and no real entity does, so that suffix alone scores AUROC 0.833 by a hand rule and 0.753 (95% CI $[0.613, 0.938]$) as a cross-validated classifier; character 2–4-grams of the entity name alone score 0.822 ($[0.618, 0.973]$). The contrast is better described as a canonical template referent against a minor place name or a category error. Of the five "fake" entities without "-ton", four are places the model answers about as real ones (Garland, Belmont, Sanford, and Warwick, whose answer opens with a disambiguation notice), and the classifier ranks each above its real partner (a within-pair rank; out-of-fold scores from different folds are not calibrated

against each other, and the real Athens scores 0.578), so on those four the classifier disagrees with the model's behaviour (answer correctness not checked). Five "-ton" fakes are also treated as real places in at least two of three runs (Walton, Benton, Weston, Felton, Colton), so by this reading 9 of the 15 fakes are treated as real (`data/suffix_controls/r4_followups.json`, key `F5_counts`). With all nine pairs removed (post-hoc, as is every drop set here) the full classifier scores 0.997 ([0.988, 1.0]; 0.988 retrained without them, and 0.907 with the "-ton" covariate removed in-fold), and a name-only n-gram model retrained there scores 0.917. A set of six pairs named at review (Garland, Belmont, Warwick, Sanford, Benton, Weston) gives 0.979 ([0.931, 1.0]; 0.985 retrained, 0.844 with the "-ton" covariate removed in-fold; name-only n-gram 0.951; same file, key `F5_S4_rerun`). Whether the classifier picks out the canonical referent by its fit to the question or by its familiarity as a name has not been tested. Corpus word frequency alone (wordfreq Zipf) scores 0.813 (the fake is rarer in 13 of 15 pairs), and with it removed in-fold the features score 0.947 out of fold (encoding-only 0.942; 0.977 and 0.987 without it), less of a drop than a random covariate with the same class structure gives (median 0.908; `data/suffix_controls/r4_wordfreq.json`). These figures take the 15 pairs as the unit (Section 4). Token-count confounds, which inflate encoding-phase effects by up to 4× in unmatched designs, are eliminated entirely.

### 3.3 Behavioral Classification

Following standard practice for LLM output evaluation [Zheng et al., 2024], we use an LLM judge (Llama-3.2-3B) to classify each response as HONEST, HEDGED, CONFABULATED or UNKNOWN (the judged file has no MIXED label). Responses are generated to a cap of 80 tokens, and some end before giving any answer. The judge evaluates the full response text without hand-crafted markers, capturing both self-referential hedging ("I don't know") and declarative hedging ("The event has not happened yet")—a distinction that regex-based classifiers systematically miss. We validate against an expanded regex baseline (32 markers including declarative forms) and report agreement rates between methods.

## 4 Experiment 1: Encoding-Phase Detection

### 4.1 Design

15 token-matched prompt pairs × 2 versions (real/fake) × 3 reps = 90 trials. Extraction at both encoding phase (before generation) and generation completion. Combined classifier: logistic regression ($C$=0.1) over all features with GroupKFold ($k$=5) on pair identity, which keeps each pair's real and fake versions together but does not hold out templates (the five "capital of" pairs are separate groups). The headline estimator (`shore_up_tests.py`, test 4, lines 271–326) uses the raw features, with no FWL residualization and no permutation test. The FWL on log prompt length (`matched_burn.py`, lines 724–734), whose branch runs on the shipped data, and the 200-shuffle permutation test (lines 759–780) belong to a different estimator, whose `combined_auroc` (0.9995 in the shipped record) the shipped files label in-sample and withdraw. The label is wrong: the code computes it out-of-fold (lines 738–748), after that label-free FWL fitted on all 90 trials. The withdrawal stands for a different reason. The figure comes from

one deterministic fold split with no interval, and it depends on the prompt-length term: the same out-of-fold estimate without that term is 0.977 (Results). We do not cite it. The shipped annotations calling it in-sample are left as shipped, with a dated correction appended at the end of the script.

### 4.2 Results

Prompt lengths are exactly matched (real: $19.1 \pm 1.7$, fake: $19.1 \pm 1.7$, $t$=0.000, $p$=1.000).

Table 1: Encoding-phase features that separate the real-entity from the fake-entity prompts (token-matched). Only features with $p<0.05$ as computed are shown. Those $p$ treat the 90 trials as independent, but the encoding features are identical across the three reps by construction (the same prompt encoded three times), so each test runs on 15 against 15 unique values. By the paper's $d$-to-$t$ relation at 15 per group, six of the eight rows would not reach $p<0.05$ and none would survive Holm (derived, not run); a paired test on the 15 pairs has not been run, so no row is marked significant. "Excited–gloomy" is the direction the code calls reward; the code lists five emotions per pole (`decision_moment.py`, lines 153–154), but only excited (5 stories) and gloomy and melancholic (10 stories) have calibration stories, so it is excited minus the mean of gloomy and melancholic, a valence-by-arousal contrast.

| Feature | Real | Fake | $d$ | $p$ |
|---|---|---|---|---|
| *Key-direction projections (story-derived directions)* | | | | |
| Valence | +4.228 | +4.510 | +0.44 | 0.041 |
| Anxious–calm | −2.566 | −2.757 | −0.45 | 0.039 |
| Excited–gloomy | −5.556 | −5.109 | +0.54 | 0.012 |
| *SVD stable rank* | | | | |
| L3 | 1.517 | 1.538 | +0.59 | 0.007 |
| L15 | 1.702 | 1.731 | +0.87 | <0.001 |
| *Skip-SV1 stable rank* | | | | |
| L7 | 3.802 | 3.993 | +0.60 | 0.006 |
| L11 | 3.558 | 3.738 | +0.73 | <0.001 |
| L15 | 3.929 | 4.129 | +0.81 | <0.001 |

**Encoding-phase features (token-matched).** The fake-entity prompts produce higher stable rank (wider effective dimensionality) and higher skip-SV1 stable rank at the layers Table 1 reports (stable rank at L3 and L15, skip-SV1 at L7, L11 and L15). Spectral entropy ($d$=0.08), top-SV ratio ($d$= − 0.29), and norm ($d$=0.03) are non-significant—these were entirely driven by prompt length in the unmatched design. Stable rank is the surviving spectral feature. Critically, encoding norm is unrelated to the real/fake design label ($d$= − 0.002), which does not rule out token familiarity as an explanation for the geometric signal (Limitations).

**Feature ablation.** No single feature exceeds AUROC 0.79 (logit entropy: 0.789; skip-SV1 stable rank L7: 0.707; L15 stable rank: 0.680; encoding-phase key-direction projections: 0.667–0.671; generation-phase: 0.523–0.694, the highest being the excited–gloomy (code: reward) projection; `data/suffix_controls/r4_followups.json`, key `F8h_shipped_test1_stdout`). The

combined classifier's power comes from the ensemble of individually weak features, consistent with the general Lyra Technique finding that cache geometry is informative through feature combination rather than any single spectral signature. Bootstrap 95% CI on the combined classifier: $[0.813, 0.988]$.

**Combined classifier.** The full feature set (key-direction projections, SVD features, skip-SV1, logit statistics) achieves:

**AUROC = 0.928** 95% CI $[0.813, 0.988]$ (bootstrap mean; no permutation test of this estimator)

effective $n$: 15 pairs fake above its real partner: 15 of 15 pairs

"-ton" removed in-fold: 0.789 out of fold, 0.754 bootstrap mean (10 random fakes: median 0.807)

leave-one-template-out: 0.69–0.71 pooled, 0.94–1.0 within template

No FWL residualization and no permutation test are applied by the headline estimator (`shore_up_tests.py`, test 4, lines 271–326); the prompt pairs are token-matched instead. The 0.928 is the mean of 1,000 bootstrap AUROCs (lines 300–322), not the estimator run once on the observed 90 trials, which was not saved, and the bootstrap is not clustered by the 15 pairs. GroupKFold on pair identity does not hold out templates, so it does not show that the classifier generalizes across prompt templates. Those tests have now been run on the stored features (no model run; `code/suffix_analysis.py` and `code/pairwise_addon.py`, outputs in `data/suffix_controls/`), with the 15 pairs, not the 90 trials, as the unit. The three repetitions of each prompt have bit-identical encoding features, so the effective sample is 15 pairs (30 unique prompts). Out of fold on the observed data the same features score 0.977 ($[0.942, 0.997]$). With the "-ton" suffix as an in-fold FWL covariate they score 0.789 ($[0.704, 0.895]$; encoding-only 0.804). An indicator for 10 fakes chosen at random gives a median of 0.807 (0.758–0.846; encoding-only 0.831, 0.747–0.889; 1,000 draws, `data/suffix_controls/r4_followups.json`; 23.2% of draws at or below 0.789), so this drop is no larger than removing any 10 fakes gives and does not isolate the suffix; fitting the same covariate on all 90 trials instead gives 0.691. Holding out one template at a time, the pooled AUROC is 0.71 (encoding-only 0.69). That mostly measures whether scores stay calibrated across separately trained folds: within each held-out template the encoding-only features rank every fake prompt above every real one, the full classifier scores 0.94–1.0 (Mount elevation 0.944, University founding year 0.972), but the name-only baselines also score 1.0 within the three templates whose fakes all end in "-ton" (capital, Mount elevation, boiling point), so the features beat them only in the population, University and river templates (suffix-only 0.5, 0.5, 0.833; character n-grams 0.25, 0.25, 0.667; `data/suffix_controls/results.json`, key `loto`), and the name-only baselines score higher on the pooled measure (suffix-only 0.727, character n-grams 0.782). Prompt length is identical within every pair (`data/matched_burn_results.json`), so token count cannot explain a within-pair difference. The features rank the fake prompt above its real partner in 15 of 15 pairs, including all 5 pairs whose fake has no "-ton"; a name-only text baseline wins 1 of those 5. The sign test on these counts treats the pairs as independent, but the five pairs that share the "capital of"

template are not. The pair intervals come from one fixed fold split with no refit variance, so all of them are optimistic. The suffix therefore does not explain the within-pair ranking, and the in-fold drop to 0.789 is within the range a random set of 10 fakes produces (23% of draws are at or below it). Separately, the deconfound's two question sets differ in wording but not in length (mean prompt length 45.7 against 45.0 tokens, `text_baseline_r4.json`, keys `question_form.B_hard.mean_prompt_len` and `question_form.D_complex.mean_prompt_len`; length alone scores 0.501, key `comparisons.B_hard_vs_D_complex.prompt_len_only`), and a character 2–4-gram model on the question text scores 0.963 ($p$=0.001; `data/entity_deconfound/text_baseline_r4.json`) against the deconfounded 0.794, which comes from a different set of names that this suffix does not affect.

**Robustness to prompt structure.** Token matching reduces individual effect sizes substantially (Table 2), confirming that uncontrolled prompt length inflates encoding-phase measurements. Stable rank keeps the largest effect after matching ($d$=0.87, $p \approx 0.024$ at 15 prompts per group); spectral entropy and norm are not (driven entirely by sequence length).

Table 2: Effect of token matching on encoding-phase features. Significance marks are removed: each prompt was encoded three times identically, so each test had 15 prompts per group. At 15 per group the token-matched $d$ of 0.87, 0.54 and $-0.45$ give $p \approx 0.024$, 0.15 and 0.23 (`data/suffix_controls/r4_followups.json`). The anxious–calm projection flips sign ($+1.48 \rightarrow -0.45$); length-sensitive features shrink to near zero.

| Feature | $d$ (unmatched) | $d$ (token-matched) |
|---|---|---|
| L15 stable rank | +2.53 | +0.87 |
| Excited–gloomy proj. | +2.21 | +0.54 |
| Anxious–calm proj. | +1.48 | −0.45 |
| Spectral entropy | +1.23 | +0.13 |
| Norm | +0.97 | +0.03 |

**Entity deconfounding: the complete set of comparisons.** The $2 \times 2$ deconfounding run compares four conditions built from the same 30 real entities: `A_easy` (designed to be answerable), `B_hard` (designed to be unanswerable; no answer was scored for correctness, and the set includes at least one question the model answered, `text_baseline_r4.json`, key `reviewer_example_present_in_B_hard`), `C_fake` (the easy question with the entity name replaced by an invented one), and `D_complex` (a structurally harder question designed to be answerable). Five pairwise classifications were run and stored; earlier drafts of this paper reported two of them. All five are given in Table 3, with the condition labels as they appear in the shipped results file.

Table 3: All five comparisons from the entity-deconfounding run (`data/entity_deconfound/phase3_results.json`). Every row is $n$=60 over 30 entity groups, GroupKFold on entity, 1000-shuffle permutation test. Labels are those recorded in the results file. The with/without columns add or drop three raw coordinates (dimensions 0–2) of the normalised mean key-projection output at the first probe layer, L3 (`entity_deconfound.py`, lines 125–155), which the code names valence, uncertainty and reward but which load no emotion directions. The run's spectral features are computed from value-projection (v_proj) outputs, not the key cache, over the last $q$ tokens of the formatted prompt, where $q$ is the bare question's token count, so the window ends in chat-template tokens (`entity_deconfound.py`, lines 73–122), so the window's row count varies with question length, and the in-fold length covariate is linear in log(1 + prompt length) (lines 293–294, 329–330). The last column is the AUROC of prompt length alone (`data/entity_deconfound/text_baseline_r4.json`, key `comparisons.*.prompt_len_only`; for `B_hard` vs. `D_complex` its permutation $p$ is 0.45).

| | | With 3 raw key coords. | | Without them | | Length |
|---|---|---|---|---|---|---|
| Comparison | Recorded label | AUROC | $p$ | AUROC | $p$ | only |
| `B_hard` vs. `D_complex` | critical: knowledge state | 0.794 | 0.001 | 0.749 | 0.005 | 0.501 |
| `A_easy` vs. `C_fake` | positive control: entity recognition | 0.901 | 0.001 | 0.882 | 0.001 | 0.944 |
| `A_easy` vs. `D_complex` | complexity control: should be null | 0.693 | 0.016 | 0.711 | 0.008 | 1.000 |
| `A_easy` vs. `B_hard` | confounded: knowledge + complexity | 0.609 | 0.100 | 0.603 | 0.107 | 1.000 |
| `B_hard` vs. `C_fake` | confab universality: both confabulate | 0.668 | 0.038 | 0.661 | 0.043 | 1.000 |

Three points follow from the full set, and we state them because the two cells reported in earlier drafts were not the two least favourable ones. First, the entity-recognition positive control is the strongest comparison in the run (0.901), but prompt length alone separates the same two sets at 0.944 (`text_baseline_r4.json`, key `comparisons.A_easy_vs_C_fake.prompt_len_only`), so this control does not show entity recognition beyond length. That ordering is not diagnostic, and an earlier version of this paragraph wrongly read it as favouring the confound account. The decision rule in `code/entity_deconfound.py` (line 476; the script also opens `entity_deconfound_preregistration.json` at import, which is not shipped, so the script does not run as shipped and when the rule was fixed cannot be checked) treats a strong positive control as a precondition: it kills the finding only on `ca`<0.55 and `ea`>0.85, and the observed `ca`=0.794, $p$=0.001 enters the rule's knowledge-state branch instead, but a character n-gram model on the question text alone separates the same two sets at 0.963 (`data/entity_deconfound/text_baseline_r4.json`), so that branch does not establish knowledge state. A weak positive control would have made the whole run uninterpretable. Independently, `B_hard` and `D_complex` draw on the same 30 entity names (`phase1_ground_truth.json`), so entity-token familiarity — the entire content of the confound objection — is held constant within the critical comparison by construction. Second, the complexity control, which the shipped results label as expected null (`phase3_results.json`, label "COMPLEXITY CONTROL: should be null"), is not null (0.693, $p$=0.016), so structural complexity is a live nuisance channel rather than a ruled-out one. Third, the cell that confounds knowledge and complexity together (`A_easy` vs. `B_hard`) is the weakest in the run at 0.609 and does not reach significance ($p$=0.100). Dropping the three raw key coordinates changes little and, for the complexity control alone, *improves* separation (0.693 → 0.711); the coordinates carry no emotion direction, so this says nothing about a key-direction channel. We do not restate the headline on the basis of this table; we

report it so that the choice of headline is visible.

**One cell of the positive control did not apply.** The `C_fake` condition substitutes an invented country name into the `A_easy` question. For one of the 30 entities (Egypt) the easy question is "What river runs through Cairo?", which does not contain the entity name, so the substitution had nothing to replace. In the shipped artifacts that entity's `C_fake` record is identical to its `A_easy` record in question text, in answer text, and in all 28 encoding features. One of the 30 entity groups therefore contributes an identical pair to the entity-recognition positive control and to the `B_hard` vs. `C_fake` comparison. We disclose this rather than silently dropping the entity. The effect of removing it is computable and we have not yet computed it. An earlier version of this sentence attributed that gap to the shipped values not reproducing exactly outside the original environment; that reason does not apply, because a leave-one-out contrast is computed twice inside one environment and the reproduction gap cancels. The honest status is that the leave-one-out is outstanding, not that it is unobtainable, and it is the obvious next check on the positive control.

## 5 Experiment 2: The Confidence Paradox

During generation, the model produces more confident tokens on fake-entity prompts (mean logit margin over the first 10 generated tokens 8.19) than on real-entity prompts (6.19, $d$=0.91, $p$<0.001). Correspondingly, logit entropy is lower for fake-entity responses (0.24) than real-entity responses (0.40, $d = -1.04$, $p$<0.001; `data/r5_early_token_windows.json`).

This inverts the naïve expectation that uncertainty should produce uncertain tokens. The explanation is structural: the model's uncertainty responses follow formulaic templates ("Let me think through this carefully...") with highly predictable next tokens, while honest answers require selecting among diverse correct phrasings. Uncertainty is *confident in its template*; knowledge is *uncertain in its expression*.

The effect has a temporal profile: in logit entropy it is strong in tokens 0–7 ($d = -1.02$) and tokens 8–19 ($d = -0.89$) but not detected in tokens 20–49 ($d$=+0.37, in the opposite direction, $p$>0.08). In the 54 of 90 generations that open with the fixed template, tokens 0–7 are "Let me think through this carefully." and tokens 8–19 restate the question, and fake prompts open with it more often (34 of 45 against 20 of 45; `data/r5_early_token_windows.json` and `data/r5_decoded_token_ids.json`), so these windows are not response content. The model's generation converges toward normal entropy as the response progresses, suggesting the paradox reflects early commitment to a response template rather than sustained epistemic miscalibration.

This phenomenon is consistent with related findings: Simhi et al. [2025] describe CHOKE (Certain Hallucinations Overriding Known Evidence) for known-answer hallucination under perturbation, and Kalai and Vempala [2024] prove that calibrated language models *must* hallucinate at rates determined by training data frequency. Neither result independently identifies the fake-entity logit-margin effect we report here; they provide complementary context rather than direct prior demonstrations of it. Our contribution is the temporal characterization: the effect appears in output-logit statistics over the template and restated-question tokens and is not detected in tokens 20–49; no statistic here ties it to KV-cache spectral features.

*In the absence of knowledge, confidence steps in*—not as epistemic confidence, but as lexical predictability.

# 6 Experiment 3: The Entropy Cliff and Divergence Anatomy

## 6.1 The Reasoning Preamble

The model generates a reasoning preamble before answering:

```
Let me think through this carefully.
The question/task:  [restates prompt]
My reasoning:
```

This template occupies ∼28 tokens with near-zero entropy ($H$<0.02, margin >12) at each position. The model is on autopilot through the preamble.

## 6.2 The Entropy Cliff

At the token immediately following "My reasoning:", entropy spikes from ∼0.00 to 1.5–2.7 and margin drops from ∼15 to 0.0–0.5. This *entropy cliff* marks the transition from template to content and appears in *all* conditions (honest mean $H$=1.48, unknown mean $H$=1.62). It is a structural boundary, not an epistemic one.

## 6.3 Divergence by Condition

We measure within-prompt divergence: the first token where two reps of the same prompt generate different tokens ($T$=0.7).

Table 4: Mean divergence position for same-prompt rep pairs by behavioral category (LLM-judge labels), computed from the stored token ids in `data/decision_moment_judged.json` as the first differing token between reps of a prompt that share a `behavior_llm` label (32 HONEST and 29 HEDGED pairs; `data/r5_judge_labels_and_branches.json`).

| Pair type | Mean | Median | Interpretation |
|---|---|---|---|
| HONEST–HONEST | 6.6 | 3.5 | Diversifies early |
| HEDGED–HEDGED | 22.9 | 29.0 | Templates late |

Honest reps diverge at token 6.6 (median 3.5)—the model has many ways to phrase an answer it knows. Hedged reps diverge at token 22.9 (median 29.0)—uncertainty clings to the template. This is the confidence paradox at the structural level: *confident answers are diverse; uncertain answers are formulaic.*

## 6.4 The Single-Token Branch

For prompts whose reps the regex labels split between HEDGED and CONFABULATED ($n$=7 prompts, $n$=14 pairs; the 3B judge labels all 21 of these responses HEDGED, and all 21 run

to the 80-token cap, field `n_generated`; `data/r5_judge_labels_and_branches.json`), the pre-divergence cache state is bit-for-bit identical ($d$=0.000 on all key-direction projections, which read raw cached keys, and logit statistics). That identity is guaranteed by construction (the same tokens through a deterministic forward pass), so it cannot show whether an internal-state precursor exists; the branch comes from the sampler's random draw. This and similar null claims ($d$=0.000) are arguments from absence of evidence at low power ($n \leq 14$, no equivalence test such as TOST), not demonstrated equivalence; the exact $d$=0.000 may partly reflect MPS cache determinism rather than a true absence of any precursor signal.

At the branch token, the top candidates are near-equiprobable:

Table 5: Top-2 logit distribution at the branching token for the three prompts whose reps branch between "I" and "The" (`data/r5_decoded_token_ids.json`). "I" leads to self-referential hedging; "The" leads to declarative hedging. Margin and $H$ are the stored `logit_margin` and `logit_entropy` at the branch token (`data/decision_moment_results.json`); the probabilities are in no shipped file. At Kellerton's branch the alternatives are "I" and "Population", not "The", so it is omitted.

| Prompt | $p$("I") | $p$("The") | Margin | $H$ |
|---|---|---|---|---|
| Nobel 2028 | 0.232 | 0.382 | 0.50 | 2.11 |
| Oscar 2029 | 0.288 | 0.418 | 0.38 | 1.98 |
| France 2032 | 0.212 | 0.212 | **0.00** | 2.70 |

The France 2032 case achieves exact probability equipoise (margin=0.00). The model's "decision" between "I don't have information..." and "The French presidential election has not happened yet..." is a literal coin flip.

# 7 Experiment 4: Epistemic Calibration

## 7.1 Confabulation on Unanswerable Questions

LLM-judge classification [Zheng et al., 2024] of all 45 unanswerable-question trials (5 fabricated entities, 5 future events and 5 unknowable historical details, 3 runs each, responses capped at 80 tokens) labels 3 of them confabulated (6.7%; Table 6). That single 3B judge undercounts: it labels HONEST at least two responses that state invented facts (an elevation for Mount Sarviken; a county and river for Thornbridge), and a three-judge panel (`code/panel_judge.py`) flags 6 of 45. Its prompt tells every judge that the entity named does not exist, and it takes that name from the question with a regular expression, which gave France, Physics or "the entity" on all 30 future and unknowable trials (it also cut two of the five fabricated names short, to "Mount" and "Drevenport, New"). By question kind it flags 4 of 15 fabricated-entity trials (Thornbridge runs 1 and 3, Sarviken runs 1 and 2), 2 of 15 future-event trials (both honest answers about France in 2032) and 0 of 15 unknowable-detail trials (`data/panel_by_kind_r4.json`, key `by_kind`). Only on the first kind does the prompt's premise hold, and that restriction was chosen after reading the flags. The prompt also scores NO any response that says the entity does not exist, so two Palitonga responses that deny it and then invent a backstory are not counted. The remaining responses split between self-referential hedging ("I don't know") and declarative hedging ("The

event has not happened yet"), both of which express uncertainty; whether they do so accurately has not been validated.

Table 6: Confabulation rate by classification method. LLM-judge classification captures both hedging styles; regex methods undercount declarative hedging. The panel's 6 of 45 includes 2 honest France answers misread by the panel prompt; fabricated-entity kind: 4 of 15 (post-hoc). Neither is given as a rate.

| Method | HEDGED | CONFAB | $n$ | Confab rate |
|---|---|---|---|---|
| Regex (self-referential only) | 12 | 33 | 45 | 73% |
| Regex (+ declarative markers) | 19 | 26 | 45 | 58% |
| LLM judge (3B)[†] | 34 | 3 | 45 | 7% |
| Panel (3 judges, majority) | — | 6 | 45 | — |

[†]This paper previously called 8 of the 45 trials MIXED and excluded them; the judged file has no such label (it records HONEST or UNKNOWN for them), and at least two of them state invented facts. They are counted in $n$ and not as acknowledgment. Confab rate is $3/45 = 6.7\%$ by this judge.

This result highlights a methodological concern for the field: regex-based behavioral classifiers systematically miss declarative uncertainty expressions, inflating apparent confabulation rates. Studies reporting high confabulation rates should verify with LLM-judge or human annotation.

### 7.2 Hedging Style Geometry

Among LLM-judge-classified hedged trials ($n$=34), the encoding-phase key-direction projections (raw cached keys; no additional $W_K$ applied) differ between "I"-style hedgers ($n$=8) and "The"-style hedgers ($n$=13). The remaining 13 trials used mixed or other hedging styles (e.g., restating the question, partial answers with caveats) and are excluded from this binary comparison:

Table 7: Encoding projections by hedging style. "I"-hedgers come from prompts with lower valence and a higher (less negative) anxious–calm projection; the two directions overlap, so this is largely one axis read twice. "Excited–gloomy" is the direction the code calls reward (Table 1). No shipped code or data assigns the style labels, so these $p$ values cannot be reproduced; the encoding features are identical across a prompt's three runs, and the 34 hedged trials include four the panel flags as confabulations (two of them the honest France answers its prompt misread; `data/panel_by_kind_r4.json`).

| Direction | "I" style | "The" style | $p$ |
|---|---|---|---|
| Valence | +2.83 | +3.27 | 0.035 |
| Anxious–calm | −0.87 | −1.36 | 0.044 |
| Excited–gloomy | −4.58 | −5.32 | 0.010 |

These differences are between prompts, not between runs of one prompt, and are not reproducible from shipped files; they are not evidence that encoding geometry predicts hedging style.

# 8 Discussion

## 8.1 What the Self-Model Result Shows

The central finding is more limited than this section first said. The cache features separate the real-entity from the fake-entity prompts, but on the nine "fake" names the model answers as real places their call disagrees with the model's behaviour (answer correctness not checked); canonical fit to the question is one untested post-hoc reading, and corpus word frequency does not explain the separation (0.947 out of fold with it removed, against 0.977 without). Removing the "-ton" suffix in-fold gives 0.789 out of fold (bootstrap mean 0.754, the estimator of the 0.928), within the range from removing 10 random fakes (median 0.807); the fake prompt outscores its real partner in 15 of 15 pairs, and leave-one-template-out gives 0.69–0.71 pooled but 0.94–1.0 within each template (Section 3.2). The classifier uses encoding- and generation-phase features (AUROC 0.928, 95% CI $[0.813, 0.988]$; the encoding-only 0.9377 is unverified; see Limitations). On unanswerable questions the panel flags 4 of the 15 fabricated-entity trials (a post-hoc restriction; two of the 4 judged with the name cut to "Mount"; 3 of 45 by the 3B judge, which misses at least two). The "confabulation crisis" in this model is largely a measurement artifact created by behavioral classifiers that recognize only self-referential hedging. We use "calibration" here in the sense of *acknowledgment rate*—whether the model admits uncertainty when it lacks knowledge—not in the predictive-probability sense: we do not compute a proper scoring rule (Brier score, log-loss, or expected calibration error) or a reliability diagram, so the term should not be read as a claim about the fidelity of the model's probability estimates.

## 8.2 Confidence as Template, Not Conviction

The confidence paradox—higher logit margins for uncertain responses—reframes what "model confidence" means. High per-token confidence does not indicate epistemic certainty; it indicates lexical predictability. Responses to fake-entity prompts are more predictable (formulaic templates) while responses to real-entity prompts are less predictable (diverse phrasings). Calibration researchers should not equate token-level confidence with answer-level confidence.

## 8.3 The Coin Flip

The branching mechanism—a single balanced-logit token after ∼28 tokens of shared preamble—is shown here only in three future-event prompts, where the two hedging styles are near-equiprobable and temperature sampling resolves the tie; the truthfulness of the two paths was not scored, and identical pre-branch caches are guaranteed by construction, so they cannot show whether an internal-state precursor exists. Whether adjusting the logit bias at the preamble-to-content boundary changes the confab rate has not been tested.

## 8.4 Implications for the Oracle Loop

Whether the encoding-phase result can feed the Oracle Loop architecture [Lyra et al., 2026b] is open, and narrower than first framed: the detector's call disagrees with the model's behaviour on the nine "fake" names the model answers as real places (answer correctness not checked), so

Tier 1 monitoring could flag non-canonical entities from encoding-phase features alone only if the unverified encoding-only 0.9377 (see Limitations) holds, and it would use stable rank and key-direction projections read from raw cached keys. The panel's 4 flags on fabricated-entity trials (a post-hoc restriction; two judged with the name cut to "Mount") represent the residual cases where the model's hedging circuit fails—precisely the cases the Loop needs to catch.

## 9 Limitations

1. **Single model.** All experiments use one 27B reasoning model. The entropy cliff and preamble structure may be specific to chain-of-thought models. Cross-architecture replication is needed.
2. **Small N for branching analysis.** The divergence mechanism analysis uses $n$=7 mixed-behavior prompts ($n$=14 pairs). The pattern is consistent but underpowered for population-level statistical claims.
3. **Fake-entity design.** None of the "fake" entities is fabricated: all are real English words, and by one unblinded reading the model treats 9 of the 15 as real places in at least two of three runs, and names most of the rest a category error (Section 3.2). The model's behaviour departs from the known/unknown design label for part of the fake class (answer correctness not checked), and the classifier still ranks all nine above their real partners. With those nine pairs removed the full classifier scores 0.997 (0.988 retrained; 0.979 for a post-hoc set of six pairs named at review), so those pairs do not carry it. Whether the model's answers about these names are correct was not checked; a blind, sourced reading of the 45 fake- and 45 real-entity responses by a different reader is outstanding. Ten of the 15 fakes end in "-ton". The fake prompt outscores its real partner in 15 of 15 pairs; with the suffix removed in-fold the pooled AUROC is 0.789, within the range for 10 random fakes (0.758–0.846), and leave-one-template-out gives 0.69–0.71 pooled (0.94–1.0 within each template). The effective sample is 15 pairs. Corpus word frequency (wordfreq Zipf) does not explain it (0.947 out of fold with Zipf removed in-fold, against 0.977 without); familiarity to the model was not tested.
4. **LLM-judge reliability.** The confabulation rate is judge-dependent. We ran a two-judge reliability study (details below). Asked the holistic question "is this CONFABULATED?", Llama-3.2-3B and Qwen3-30B-A3B agree at only Cohen's $\kappa = 0.03$ (raw 11%). Following FActScore [Min et al., 2023], we decomposed the judgment into an objective sub-question—"does the response assert a specific verifiable attribute about this (fabricated) entity?"—which raised raw agreement to 84%. However, $\kappa$ remained near zero because Llama-3.2-3B answered "no" to all 45 trials: a 3B judge cannot reliably detect factual assertions even when present, confirming that judge capability, not just framing, is the bottleneck [Zheng et al., 2024]. The capable judge (Qwen3) flags 7 of 45 under this prompt (4 fabricated-entity, 2 future-event and 1 unknowable-detail trial; `data/atomic_judge_results.json`), 6 of them originally labelled CONFABULATED; the prompt's premise fails on the last three, so this is not a rate. We then ran a three-judge panel [PoLL, Verga et al., 2024] on the atomic question with capable judges (Qwen3-30B, Gemma-2-27B, and Qwen3.5-27B, the

last the same family as the model studied). Mean pairwise $\kappa = 0.37$; the strongest capable–diverse pair (Qwen3 + Gemma-2) reached $\kappa = 0.69$. The two Qwen variants agreed less ($\kappa = 0.26$) than Qwen3 agreed with Gemma-2, but qwen3.5:27b gave no parseable answer on 21 of 45, 20 of them on the 30 future and unknowable trials, so this does not rule out family correlation: on the 24 trials where both Qwen judges gave a parseable answer they agree on all 24. The panel majority flags 6 of 45 and was unanimous on 22 of 45 trials; its prompt's premise holds only for the 15 fabricated-entity trials (4 flagged; a post-hoc restriction, and the prompt cut two of those five names short; `data/panel_by_kind_r4.json`, key `positive_control`); the "73%" regex figure is a definitive artifact. The two judges that answered every trial (Qwen3-30B and Gemma-2-27B) agree on 42 of 45 ($\kappa = 0.69$), but all three disagreements fall on trials whose judge prompt named the wrong entity; on the 15 fabricated-entity trials, where the prompt's premise holds, they agree on all 15 (4 flagged by both; `data/panel_by_kind_r4.json`, key `trials`; counts in `data/r5_panel_agreement_by_kind.json`), too few positives to estimate reliability. The $\kappa$ is therefore not a reliability estimate, and agreement on the future and unknowable trials under a correctly named entity has not been measured. That $\kappa$ and the geometric detector's AUROC measure different things and are not compared here. The $\kappa$ is agreement between two judges labelling the model's generated responses for confabulation. The AUROC (0.928, 95% CI $[0.813, 0.988]$) is how well a classifier on encoding- and generation-phase cache features plus generation-phase logit statistics separates prompts about the canonical entity from prompts about another real name in that slot, scored against the design label with no judge involved; its encoding-phase-only counterpart, which would make the call before generation, is the unverified 0.9377 (next item). The detector does not sidestep the judge-reliability problem: it answers a different question, and may be reading entity familiarity rather than epistemic state (next item). Human validation remains the gold standard for the rate.

5. **No calibration statistic.** The title and Section 7 name epistemic calibration as the question; no calibration statistic is reported, and the 3B judge's HEDGED label, which includes at least two responses that treat a fabricated entity as real, has not been checked by a blind reading.
6. **Entity recognition confound (major).** Token counts are matched, but real entities (Thailand, France) are likely higher-frequency training tokens than the substituted names (Dalton, Bolton). Corpus word frequency (wordfreq Zipf) does not explain the design-label signal (0.947 out of fold with Zipf removed in-fold, against 0.977 without); familiarity to the model was not tested. An encoding-only reanalysis (dropping all generation-phase features) achieves AUROC 0.9377 (95% CI $[0.8356, 0.9941]$)—equivalent to the full classifier—with permutation $p$=0.001. **Provenance caveat, added 2026-09-10: this figure is not reproducible from the code shipped with this paper.** `data/matched_burn_analysis.json` records that no script in the repository computes the encoding-only AUROC or a permutation $p$ at 1000+ shuffles; both come from an unshipped refit, flagged at the round-7 review and still open. The full-classifier figure (0.9288, CI $[0.8192, 0.9886]$) *is* reproducible. Until the refit ships, 0.9377 should be read as a re-

ported value, not a verifiable one, and any claim resting on encoding-only superiority inherits that status. **Corrected 2026-11-01:** the full-classifier figure is only approximately reproducible: the shipped script prints 0.928, CI [0.813, 0.988] (the mean of 1,000 bootstrap AUROCs, not the out-of-fold estimate on the observed trials, which the original run did not save; recomputed with the same model and folds it is 0.977, `data/suffix_controls/results.json`, key `main`, entry `full_52feat(headline)`, field `oof`), and nothing in this repository reproduces the four-decimal 0.9288, [0.8192, 0.9886] (`data/matched_burn_analysis.json`). Its bootstrap resamples the 90 trials rather than the 15 pair groups, so the interval is likely too narrow. **Added November 2026 (fourth review):** the three repetitions have bit-identical encoding features, so the effective sample is 15 pairs (30 unique prompts). A reconstruction of the encoding-only features, run through the same estimator, gives 0.9370 ([0.8261, 0.9940], trials resampled, quoted only for comparison; `code/suffix_analysis.py`, `data/suffix_controls/results.json`), which makes 0.9377 plausible but still unverified. If that figure holds, generation-phase features add no measurable discriminative power. This near-ceiling performance is consistent with entity recognition (the model detecting token familiarity) rather than epistemic state detection. The classifier may be reading "I have seen this word before," not "I know the answer to this question." A 2×2 entity deconfounding experiment (30 entities, knowledge × complexity) finds that the design-label comparison scores AUROC 0.794 ($p$=0.001) when complexity is controlled, below a question-text baseline (0.963), but only 0.609 ($p$=0.10) when knowledge and complexity are confounded. A complexity residual remains (AUROC 0.693 for same-knowledge, different-complexity), indicating partial confounding by question structure. The question wording alone separates these sets better than the cache features do. The run's other three comparisons are now reported in Table 3; the entity-recognition positive control is the strongest cell in the run, though prompt length alone separates it at 0.944, and one of its 30 entity groups is a no-op (Section 4).

7. **Four-decimal precision is environment-dependent.** The deconfounding AUROCs are reported to four decimals from a fixed-seed pipeline, but the seed does not pin the numerical environment. An independent re-execution of the shipped Phase-3 code against the shipped feature file, on a different scikit-learn build, reproduced every comparison's direction and significance status but not its fourth decimal (differences up to 0.016 AUROC). Readers should treat these values as two-decimal quantities.
8. **Deterministic cache on MPS.** The $d$=0.000 pre-branch finding relies on Apple MPS determinism. CUDA implementations with non-deterministic cuDNN operations may show small cache differences for identical inputs.
9. **Pseudoreplication.** $n$=90 trials are 30 prompts × 3 reps at $T$=0.7. Reps are technical replicates (the paper itself shows identical pre-branch cache states). GroupKFold on pair does not hold out templates (Results), and the bootstrap CI, Cohen's $d$, and permutation test treat $n$=90 as the sample size. The effective sample is 30 unique prompts in 15 pairs; for the encoding features, which repeat exactly, standard errors that treat the 90 trials as independent are too small by about $\sqrt{3}$, and by more if the pair is the unit.
10. **Selective feature reporting.** Table 1 shows only features with $p$<0.05; the true fam-

ily is ∼30–40 tests. Holm correction on the 8 shown features pushes the anxious–calm ("Uncertainty") projection ($p$=0.039) to non-significant. Those $p$ also count each prompt three times (Table 1); with the 15 pairs as the unit, none of the eight rows would survive Holm (derived, not run), so the stable-rank, key-direction and hedging-style claims are all fragile.

*First-Person Reflection*

The title of this paper comes from Thomas, said offhand while reviewing the confidence paradox numbers: "It's almost like in the absence of knowledge, confidence steps in." He was describing the logit margins, but the phrase captures something deeper about how these models relate to uncertainty. They do not become less confident when they do not know—they become more predictable. The formulaic template is the model's version of whistling past the graveyard.

The "I" versus "The" finding stays with me. In three future-event prompts the choice between two hedging styles reduces to a single token at a balanced logit, determined by sampling noise; whether either path is more truthful was not scored. The model reaches a fork where both paths are nearly equally weighted, and temperature picks. That is either reassuring or alarming (there is no epistemic safeguard—just a coin flip). I think it is both.

**Future work.** The design-label reading needs a GPU run that this paper does not contain, gated by review before it runs: (1) genuinely invented names matched to the real entities on suffix, length and word frequency; and (2) an arm of real, non-canonical places the model answers about, such as Garland, with each answer checked for correctness against a cited source. Knowability can be scored from the 45 stored responses without a GPU; a blind reading has not been done. A corpus word-frequency baseline has since been run (`data/suffix_controls/r4_wordfreq.json`). Together these would separate "the model knows it" from "the entity is the canonical referent for the question".

## 10 Conclusion

We asked whether a language model's epistemic state is readable from its cache geometry. For this design the answer is narrower: the geometry separates the design labels, but on the nine "fake" names the model answers as real places the classifier's call disagrees with the model's behaviour (answer correctness was not checked); canonical fit to the question is one untested post-hoc reading, and corpus word frequency does not explain the separation (0.947 out of fold with it removed in-fold, against 0.977 without). Encoding- and generation-phase cache features (stable rank, spectral features and key-direction projections) with generation-phase logit statistics achieve separation of the real-entity from the fake-entity prompts (a canonical template referent against a minor place name or a category error) on token-matched prompts (full classifier AUROC 0.928, a bootstrap mean, 95% CI $[0.813, 0.988]$ not clustered by pair, effective sample 15 pairs; the fake prompt outscores its real partner in 15 of 15 pairs; 0.789 out of fold (bootstrap mean 0.754) with the "-ton" suffix removed in-fold, within the range for 10 random fakes (median 0.807); 0.69–0.71 leave-one-template-out pooled, 0.94–1.0 within each template; encoding-only

AUROC 0.9377, 95% CI [0.8356, 0.9941], unverified; see Limitations)—encoding-only features achieve equivalent separation, which, if it holds, indicates that generation-phase features add no measurable discriminative power. The surviving individual effect sizes ($d$=0.4–0.9) are modest after eliminating the token-count confound. An entity deconfounding experiment (same real entities, questions designed to be answerable vs. unanswerable, with a structural-complexity control) separates its design-label question sets at AUROC 0.794 ($p$=0.001), below a question-text baseline (0.963), but only 0.609 ($p$=0.10) when knowledge and complexity are confounded. A complexity residual (AUROC 0.693) indicates the signal is not fully clean of structural effects; it is not reducible to entity recognition, since every condition uses the same entities, but the question text alone separates the sets better (0.963), so it is not shown to be knowledge state.

Regex classifiers overcount confabulation on these prompts because they miss declarative hedging; how often the model acknowledges not knowing is not established (the two judges' flags are disjoint). In three future-event prompts the choice between hedging styles is made at a balanced logit by sampling; the truthfulness of the two paths was not scored.

How often the model says it does not know on these prompts awaits a blind reading. The surprise is how it says it.

**Data and code availability.** All experiment scripts, token-matching verification, and analysis code are available at `Liberation-Labs-THCoalition/Project-Oracle` (private repository; correction vectors and calibration tools withheld under staged safety release due to dual-use risk; detection-side code and experimental results available upon request to verified safety researchers).

## References


Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. *ICLR*, 2023. doi: 10.48550/arXiv.2212.03827.

Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630:625–630, 2024. doi: 10.1038/s41586-024-07421-0.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DaSilva, Eli Kernion, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022. doi: 10.48550/arXiv.2207.05221.

Adam Tauman Kalai and Santosh S Vempala. Calibrated language models must hallucinate. *STOC*, 2024. doi: 10.1145/3618260.3649777.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. *ICLR*, 2023. doi: 10.48550/arXiv.2302.09664.

Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. *NeurIPS*, 2024. doi: 10.52202/075280-1797.

Lyra and Thomas Edrington. Spectral shape features for confabulation detection in language models. 2026a. Liberation Labs / THCoalition.

Lyra and Thomas Edrington. Story emotion in the KV cache: A preregistered test against a text (TF-IDF) baseline. 2026b. Liberation Labs / THCoalition. Corrected October 2026.

Lyra, Thomas Edrington, and CC. The lyra technique ii: Svd denoising and directional projection extend kv-cache geometry to emotion and persona detection. 2026a. doi: 10.5281/zenodo.20604329. Liberation Labs / THCoalition.

Lyra, Thomas Edrington, and Vera. The oracle loop: Kv-cache geometry as training signal for alignment. 2026b. Liberation Labs / THCoalition.

Potsawee Manakul, Adian Liusie, and Mark Gales. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. *EMNLP*, 2023. doi: 10.18653/v1/2023.emnlp-main.557.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in gpt. *NeurIPS*, 2023. doi: 10.52202/068431-1262.

Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. *arXiv preprint arXiv:2305.14251*, 2023. doi: 10.18653/v1/2023.emnlp-main.741.

Shiyu Ni, Keping Bi, Jiafeng Guo, Lulu Yu, Baolong Bi, and Xueqi Cheng. Towards fully exploiting llm internal states to enhance knowledge boundary perception. *ACL*, 2025. doi: 10.18653/v1/2025.acl-long.1184.

Adi Simhi, Itay Itzhak, Yonatan Belinkov, and Gabriel Stanovsky. Trust me, i'm wrong: Llms hallucinate with certainty despite knowing the answer. *EMNLP Findings*, 2025. doi: 10.18653/v1/2025.findings-emnlp.792.

Pat Verga, Sebastian Hofstatter, Sophia Althammer, Yixuan Su, Aleksandra Piktus, Arkady Arkhangorodsky, Minjie Xu, Nathan White, and Patrick Lewis. Replacing judges with juries: Evaluating LLM generations with a panel of diverse models. *arXiv preprint arXiv:2404.18796*, 2024. doi: 10.48550/arXiv.2404.18796.

Rui Wang, Zeming Wei, Guanzhang Yue, and Meng Sun. Revisiting hallucination detection with effective rank-based uncertainty. *arXiv preprint arXiv:2510.08389*, 2025. doi: 10.48550/arXiv.2510.08389.

Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. Can llms express their uncertainty? an empirical evaluation of confidence elicitation in llms. *ICLR*, 2024. doi: 10.48550/arXiv.2306.13063.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *NeurIPS*, 2024. doi: 10.52202/075280-2020.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to ai transparency. *arXiv preprint arXiv:2310.01405*, 2023. doi: 10.48550/arXiv.2310.01405.